

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

May 2014

CL307.01 Structural Analysis-III

Date: 08.05.2014, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Find plastic moment capacity of the beam shown in Figure 1.

[07]

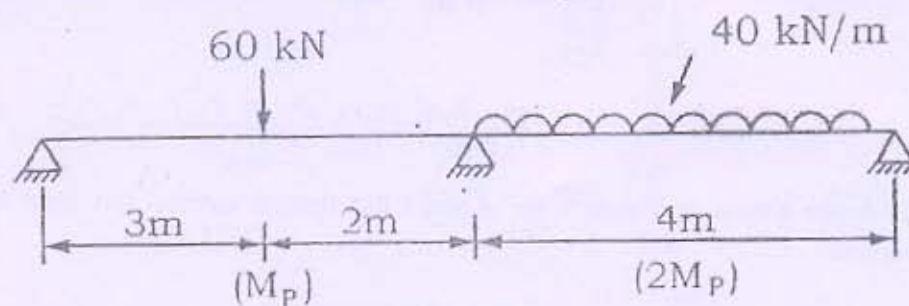


Figure 1

Q - 2 (a) Determine the shape factor for a Triangular section having base b and height h .

[04]

(b) Analyse the continuous beam shown in Figure 2 by slope deflection method and draw bending moment diagram.

[10]

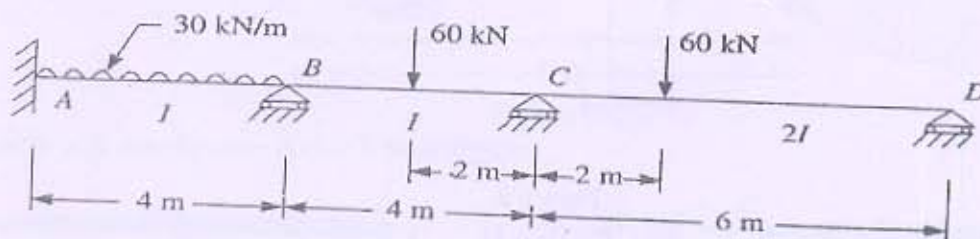


Figure 2

OR

(b) Analyse the continuous beam shown in Figure 3 by slope deflection method and draw bending moment diagram.

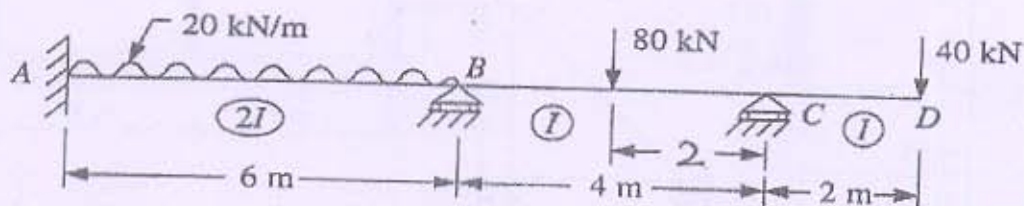


Figure 3

Q - 3 (a) Define Carry Over Moment, Carry Over Factor, Stiffness & Distribution Factor. [04]

- (b) Analyse the frame shown in Figure 4 by moment distribution method and draw bending moment diagram. [10]

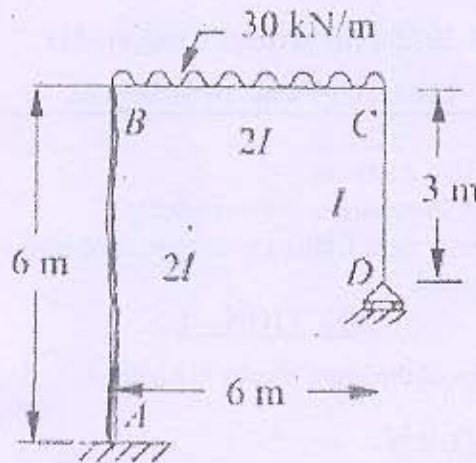


Figure 4
OR

- (b) Analyse the frame shown in Figure 5 by moment distribution method and draw bending moment diagram.

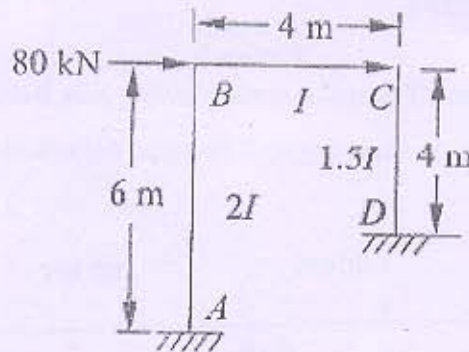


Figure 5

SECTION - II

- Q - 4 For the truss as shown in Figure 6 support A moves left by 2 mm and support B sinks by 3 mm. Take $AE = 32000$ kN, and derive load vector in structure axis. (Use stiffness method member approach) [07]

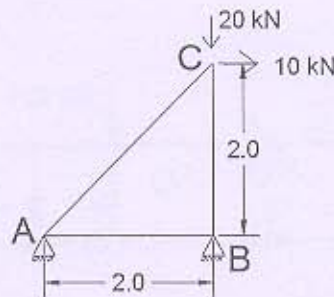


Figure 6

Q - 5

Analyse the beam as shown in Figure 7. Take EI as constant. (Use stiffness method member approach) [14]

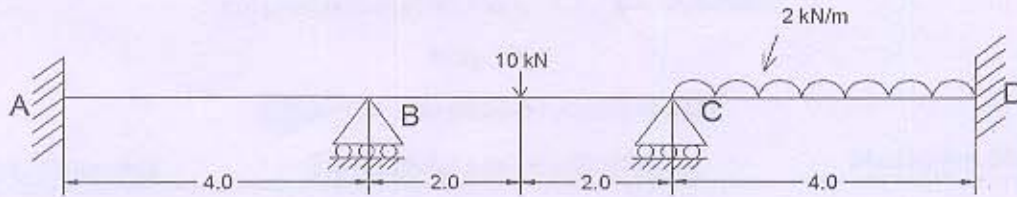


Figure 7
OR

Analyse the beam as shown in Figure 8. Take EI as constant. (Use stiffness method member approach)

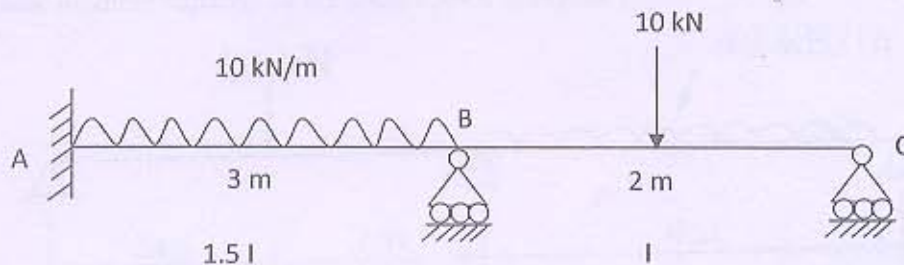


Figure 8

Q - 6 (a) Determine the fixed end moments for the fixed beam shown in Figure 9 by Column Analogy method. [05]

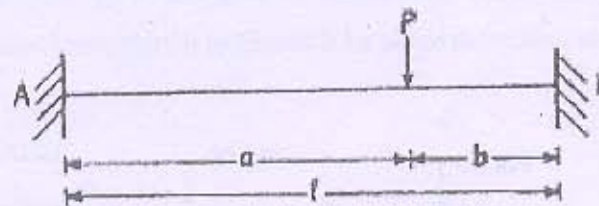


Figure 9

(b) Solve any one frame. Refer below figures.

[09]

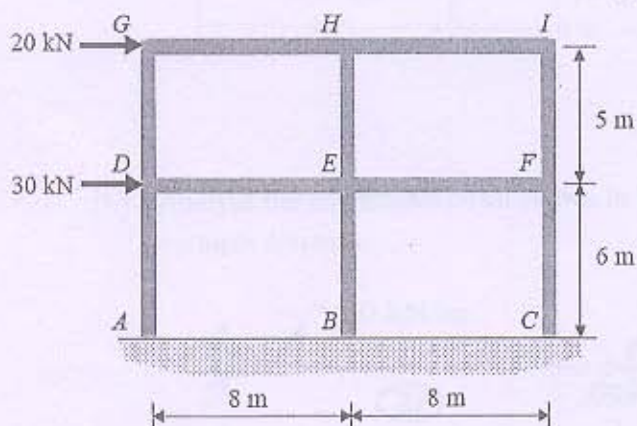
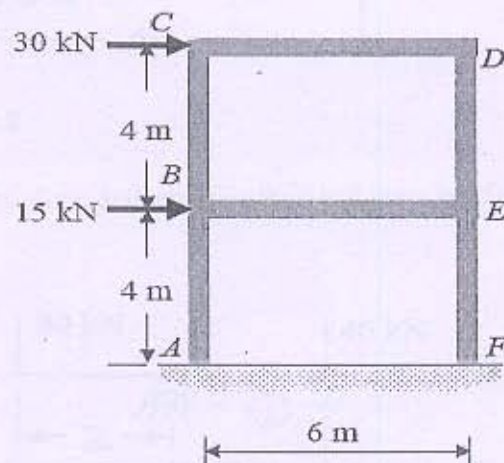


Figure 10 (Portal Method)



(Equal Area of Columns)
Figure 11 (Cantilever Method)
